Symmetry Spending Under a Quantum Shot Budget: What It Takes to Certify That Shots Were Well Spent

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Abstract

A quantum predictor that averages a model response over the images of its input under a symmetry group is a *stochastic, shot-limited object*: it applies one of several circuits and reads out a finite number of measurement outcomes. Its risk therefore depends on how its budget was spent, and two allocations with the same total number of shots can have different risks — there is no ideal limit that both are approximating. We formalise this as a *risk-at-budget* functional and decompose it into three separately measurable parts: an orbit-mean model bias, a group-draw variance that more shots cannot reduce, and a shot variance that more group draws cannot reduce. The decomposition is exact and we verify it against brute-force enumeration to 3×10^{-17} rather than asserting it. Two consequences follow. First, at fixed total shots the split between "how many symmetry images" and "how many shots per image" leaves the shot term untouched, so orbit randomisation is free in shots and the interior optimum in the split appears only when distinct circuits carry a switching cost. Second, when the *decision* about how to spend shots must itself be certified from an independent calibration experiment, the cost decomposes bound-independently as $\mathrm{Var}(D) = \mathrm{Var}_x(\mathbb{E}[D \mid x]) + \mathbb{E}[\mathrm{Var}(D \mid x)]$: at four shots per prediction the deployed-noise term exceeds the between-instance term by $15\times$ (0.07851 vs 0.00507), and at 256 shots the ordering reverses (0.00038 vs 0.00244), so shots buy the first and only preparations buy the second, down to a floor that no number of shots can cross. *Where* the certification frontier sits, however, is a property of the inequality rather than of the problem, and we report that as a withdrawal: with a distribution-free empirical-Bernstein bound, power is 0.000 up to $n = 200$ and 1.000 at $n = 400$; with a Studentised bound whose measured violation rate is 0.005 across 90 cells (nominal 0.05), power is already 0.995 at $n = 100$. At matched total budget the direction of the preparations-versus-shots tradeoff is not robust — at $B = 1600$ the allocation n=50, S=32 reaches power 0.917 while n=200, S=8 reaches 0.870, the opposite of what the loose bound suggests. We therefore claim the variance decomposition and not an allocation rule. On a bond-alternating Heisenberg response with a control-line rectification error, the same structure reproduces: full orbit averaging reduces deployed risk by 0.00800 when the target is mirror-even, and *increases* it by 0.00212 when the target is the raw response, so the value of spending shots on symmetry is governed by how much odd signal the target contains, not by whether a symmetry exists. We state explicitly which parts of this are already owned by prior work on shot/sample budget allocation and on certified symmetry release, and what remains.

1 Introduction

A growing body of work asks how a fixed quantum measurement budget should be spent. Catak [1] derives a distribution-free bound separating a sample term from a shot term and a closed-form allocation $n^\star, S^\star$ under $B = nS$. Chen [2] certifies the *release* of a symmetry constraint during training, with simultaneous confidence bounds, a post-selection oracle inequality for adaptive candidate families, and grouped-measurement shot accounting. Chen [3] treats finite-measurement inference of the symmetry group itself. Wang et al. [4] and Cai et al. [5] give randomised symmetry implementations with explicit bias–sampling tradeoffs.

This paper is about a different object, and the difference is not cosmetic.

> For a deployed predictor that draws k symmetry images of its input and spends S shots across them, the risk is a functional of the allocation. Spending the same S differently does not change how well we *know* the risk; it changes *which* risk we have.

Three things follow, and all three are measured rather than asserted.

(1) A decomposition with two invariances. The deployed risk separates as $R_S(a) = \text{bias}^2 + V_G/k + \Sigma/S$. The shot term Σ/S does not depend on k, and the draw term V_G/k does not depend on S. At fixed total shots, moving shots between "more images" and "more shots per image" leaves one term exactly fixed and moves the other. There is no interior optimum in shots alone: $k = S$ weakly dominates. The interior optimum that practitioners actually observe comes from the cost of *distinct circuits*, which is a different resource and is where a switching cost c enters.

(2) Certification cost splits into what shots can buy and what they cannot. Deciding whether symmetry spending helps requires a calibration experiment on independent labelled instances, and the deployed predictor's own shot noise enters the calibration contrast. That noise is reducible by shots; the between-instance variability of the contrast is not, and it sets a floor on $\text{Var}(D)$ that more shots leave untouched. What we do *not* claim is an allocation rule: the location of the certification frontier moves by roughly $4\times$ in n depending on which concentration bound is used, and the direction of the preparations-versus-shots tradeoff at matched budget reverses between them (§4.1).

(3) The symmetry's value is a property of the target, not of the model. On the same physical mechanism, changing only how much odd-under-mirror signal the prediction target contains flips full orbit averaging from a 0.00800 risk reduction to a 0.00212 risk increase.

2 Access model, costs, and estimands

Access. Scalar single-copy shots: one labelled state preparation per calibration instance, one circuit-execution batch per prediction, computational-basis measurement of a diagonal observable. No two-copy access, no shared Pauli measurements, no classical shadows. Every baseline receives exactly this access. Where a published method needs more — a twirl–swap test, say — we say so and do not compare it against a weaker version of itself.

Costs are counted separately. `shots` (measurement repetitions), `preparations` (independently prepared labelled states), `circuits` (distinct compiled circuits, one per symmetry image touched), and `calibration_labels`. Deployment cost per prediction is $S + c \cdot$ (distinct circuits), with c a declared stress parameter. A method that receives free calibration or free state knowledge invalidates the comparison; the policy menu and the feasibility rules below exist to make that impossible to do by accident.

Feasibility is a property of a policy, not a convention. With `per`$= \lfloor S/k \rfloor$, a policy with $k > S$ spends k shots while claiming S. This study found that bug in its own menu after a first frontier run, and the apparent power gain at small S had been bought with four times the declared shots. Every policy now carries `shots_used` and a `feasible()` predicate, the menu is filtered by it, and the dropped entries are reported.

Estimands. Primary: $\Delta_S(a) = R_S(a) - R_S(a_0)$ at matched total shots, against the raw policy a_0. Operational primary: $N_{\text{cert}}(\epsilon, \delta)$, the number of calibration preparations at which the certificate of §4 selects a policy with $\Pr[\Delta_S(a_{\text{sel}}) \le \epsilon] \ge 1 - \delta$ and power $\ge 1 - \beta$ when a useful policy exists. Fallback to a_0 is a safe abstention, not a success.

3 Risk at budget

For input x and group element g, let $\mu_g(x) = \mathbb{E}[\text{shot} \mid g, x]$ and let $\sigma_g^2(x) = 1 - \mu_g(x)^2$ for a ± 1-valued readout. A policy draws k images, spends S shots total with S/k per draw and equal weights, and outputs $Z_a(x)$. Then, exactly,

$$R_S(a) \;=\; \underbrace{\mathbb{E}_x\big[(\bar{\mu}_a(x) - y)^2\big]}_{\text{bias}^2} \;+\; \underbrace{\mathbb{E}_x\big[\text{Var}_g \mu_g(x)\big]/k}_{V_G/k} \;+\; \underbrace{\mathbb{E}_x\big[\mathbb{E}_g\, \sigma_g^2(x)\big]/S}_{\Sigma/S}, \tag{1}$$

where $\bar{\mu}_a$ is the orbit-mean response. The shot term deserves its invariance spelled out: $\sum_j (1/k)^2 \sigma_j^2/(S/k) = \bar{\sigma}^2/S$, independent of k.

Table 1: Equation (1) verified on the 2-qubit control. The residual column is the difference between the enumerated risk and the closed form; it is machine precision, not a fit.

4 Certification, and what it costs

Calibration uses n independent labelled instances; the independence unit is the instance, not the shot or the seed. For each instance we deploy the candidate and a_0 and record $D_i = (Z_a - Y)^2 - (Z_{a_0} - Y)^2$, and we bound $\mathbb{E}[D]$ with an empirical-Bernstein inequality over the fixed menu, with the multiplicity split declared in advance. The inequality is deliberately ordinary: the contribution here is not a new concentration bound but the observation that what is being bounded is a difference of two *allocation-dependent* risks of stochastic predictors, selected with the same finite-shot data that defines them — precisely the independence assumption that Catak's bound requires and states [1].

Is the transition statistics, or the inequality? The empirical-Bernstein bound has a range term $7(b-a)\ln/(3(n-1))$ that is enormous at small n whatever the data says, so a frontier located at $n \approx 400$ could be a property of the bound rather than of the problem. We therefore report $\mathrm{Var}(D) = \mathrm{Var}_x(\mathbb{E}[D \mid x]) + \mathbb{E}[\mathrm{Var}(D \mid x)]$ separately from the bound's two terms. Only the second is reducible by shots; the first is reducible only by preparations. Table 4 gives both, and the split of the bound itself.

4.1 Withdrawal: the frontier was mostly the inequality

An earlier version of this draft reported that certification is preparation-limited, on the evidence that at $B = 6400$ power jumped from 0.000 at $n \leq 200$ to 1.000 at $n \geq 800$. That claim is withdrawn. The empirical-Bernstein range term $7(b-a)\ln/(3(n-1))$ is ≈ 4 at $n = 25$ for a loss range of 4, so the bound, not the data, was deciding. Re-running the same experiment with a Studentised bound (Table 5) gives power 0.245/0.725/0.995/1.000 at $n = 25/50/100/200$ with $S = 128$, at a measured violation rate of 0.005 against a nominal 0.05 across all 90 cells — admissible empirically, though distribution-free validity is exactly what a Studentised bound does not offer, and a violation rate estimated on 400 repetitions per cell cannot resolve a true rate far below 0.05.

5 Physical tasks

Primary. Bond-alternating spin-1/2 Heisenberg chain, $n = 6$ and $n = 8$, open boundaries, quench from the Néel state, target the site-0 magnetisation. The control parameter that is odd under the chain mirror is the staggered field Δ; for even n the mirror maps $\Delta \to -\Delta$. The deployed device mis-sets that parameter by a second-order rectification of the analogue control line, $\Delta_{\mathrm{eff}} = \Delta + c\Delta^2$, whose error is odd in Δ to leading order — which is exactly the component that averaging over $\Delta \to -\Delta$ cancels. A fixed-site readout bias would *not* be cancelable by input averaging, and we say so rather than claiming credit for a symmetry that the mechanism does not have.

The three regimes differ by how much of the odd response the target contains: $y_a = (1-a)\, y_{\mathrm{even}} + a\, y_{\mathrm{raw}}$, with y_{even} the mirror-even part.

Independent evaluation. A transverse-field three-body cluster Hamiltonian at $n = 7$, a different dynamical mechanism and a different observable, with the same rectification error. The split is by task family, declared before generation.

6 What is and is not claimed

Claimed. (i) The risk-at-budget decomposition (1) with its two invariances, verified against enumeration. (ii) That under a pure shot budget orbit randomisation is free, and the interior optimum in the split is a switching-cost phenomenon. (iii) That certifying a symmetry decision is preparation-limited at fixed total budget, with the two candidate explanations of that fact measured rather than assumed. (iv) That on a physical mechanism the sign of the benefit is controlled by the target's odd content, not by the presence of a symmetry.

Regime	Policy	R_{64}	bias^2	V_G	Σ	residual
help	a0	0.03955	0.03428	0.00000	0.33686	4.2e-17
help	strat	0.00571	0.00040	0.00000	0.34004	1.0e-17
help	mix25	0.02476	0.01948	0.00000	0.33766	4.5e-17
help	mix50	0.01419	0.00890	0.00000	0.33845	2.6e-17
help	mix75	0.00784	0.00254	0.00000	0.33925	1.0e-17
help	k1	0.03949	0.00040	0.03377	0.34004	7.6e-17
help	k2	0.02260	0.00040	0.03377	0.34004	4.9e-17
help	k4	0.01416	0.00040	0.03377	0.34004	1.9e-17
help	k8	0.00993	0.00040	0.03377	0.34004	5.2e-18
help	k16	0.00782	0.00040	0.03377	0.34004	6.9e-18
help	k32	0.00677	0.00040	0.03377	0.34004	7.8e-18
help	k64	0.00624	0.00040	0.03377	0.34004	6.9e-18
hurt	a0	0.03813	0.03287	0.00000	0.33686	2.1e-17
hurt	strat	0.11438	0.10907	0.00000	0.34004	5.6e-17
hurt	mix25	0.05086	0.04558	0.00000	0.33766	3.5e-17
hurt	mix50	0.06781	0.06252	0.00000	0.33845	1.1e-16
hurt	mix75	0.08899	0.08369	0.00000	0.33925	4.2e-17
hurt	k1	0.14816	0.10907	0.03377	0.34004	1.1e-16
hurt	k2	0.13127	0.10907	0.03377	0.34004	5.6e-17
hurt	k4	0.12283	0.10907	0.03377	0.34004	1.1e-16
hurt	k8	0.11860	0.10907	0.03377	0.34004	2.8e-16
hurt	k16	0.11649	0.10907	0.03377	0.34004	1.2e-16
hurt	k32	0.11544	0.10907	0.03377	0.34004	9.7e-17
hurt	k64	0.11491	0.10907	0.03377	0.34004	1.4e-16
interior	a0	0.02071	0.01544	0.00000	0.33686	1.4e-17
interior	strat	0.01089	0.00558	0.00000	0.34004	3.5e-18
interior	mix25	0.01192	0.00665	0.00000	0.33766	2.6e-17
interior	mix50	0.00736	0.00207	0.00000	0.33845	6.9e-18
interior	mix75	0.00702	0.00171	0.00000	0.33925	1.6e-17
interior	k1	0.04467	0.00558	0.03377	0.34004	6.9e-18
interior	k2	0.02778	0.00558	0.03377	0.34004	4.2e-17
interior	k4	0.01934	0.00558	0.03377	0.34004	6.9e-18
interior	k8	0.01512	0.00558	0.03377	0.34004	2.3e-17
interior	k16	0.01300	0.00558	0.03377	0.34004	1.7e-17
interior	k32	0.01195	0.00558	0.03377	0.34004	3.5e-18
interior	k64	0.01142	0.00558	0.03377	0.34004	1.7e-17

Table 2: Risk-at-budget decomposition, enumerated exactly on the 2-qubit control. Residual is the gap between the enumerated risk and Eq. (1).

Not claimed. No new concentration inequality. No joint sample/shot allocation rule for a generalization bound — that is Catak [1]. No certified symmetry release, no post-selection oracle inequality for adaptive families, no two-copy measurement primitive — that is Chen [2]. No symmetry discovery from finite probes — that is Chen [3]. No device results, no quantum advantage, and no claim that stratified allocation $S_g \propto |w_g|\sigma_g$ is anything other than classical.

7 Limitations and what would falsify this

- Every task here is a dense statevector simulation with binomial shot sampling. The mechanisms are physical (control-line rectification, staggered-field mirror parity) but the data are not from hardware, and no device claim is made.
- The policy menu is finite and fixed before calibration. Policies generated from calibration losses would require the post-selection accounting of Chen [2], which this paper does not develop; our certificate would not cover them.

n (preps)	S (shots)	violation	power	fallback
25	256	0.000	0.000	1.000
50	128	0.000	0.000	1.000
80	80	0.000	0.000	1.000
100	64	0.000	0.000	1.000
160	40	0.000	0.000	1.000
200	32	0.000	0.000	1.000
320	20	0.000	0.140	0.860
400	16	0.000	0.976	0.024
800	8	0.000	1.000	0.000
1600	4	0.000	1.000	0.000

Table 3: Certificate power along the preparations/shots axis at fixed total budget $B = nS = 6400$, averaging-helps regime, empirical-Bernstein bound, $\epsilon = 0$, $\delta = 0.05$.

Regime	S	Var(D)	between	noise	EB var. term	EB range term
help	4	0.08354	0.00507	0.07851	0.0000	-0.007
help	16	0.01186	0.00270	0.00917	0.0133	0.014
help	64	0.00414	0.00247	0.00168	0.0196	0.205
help	256	0.00281	0.00244	0.00038	nan	nan
hurt	4	0.13272	0.01042	0.12238	0.0000	-0.007
hurt	16	0.02866	0.00766	0.02104	0.0126	0.014
hurt	64	0.01197	0.00733	0.00466	0.0200	0.205
hurt	256	0.00837	0.00727	0.00113	nan	nan
interior	4	0.07145	0.00344	0.06805	0.0000	-0.007
interior	16	0.00751	0.00086	0.00665	0.0074	0.014
interior	64	0.00166	0.00060	0.00106	0.0152	0.205
interior	256	0.00078	0.00056	0.00023	nan	nan

Table 4: What the calibration contrast is made of, and what the bound adds. Var(D) splits into a between-instance part (only preparations reduce it) and a deployed-noise part (shots reduce it). The last two columns split the empirical-Bernstein bound at the matching allocation.

- The certification frontier depends strongly on the chosen inequality — by a factor of four in n between the two bounds we tried — and the direction of the preparations-versus-shots tradeoff at matched budget is not robust to that choice (§4.1). The bound-independent content is the variance decomposition, not a frontier location.
- On the independent-evaluation task the two target variants ($a = 0$ and $a = 1$) give identical risk to five decimals, i.e. that task's response is nearly even in Δ and does not exercise the odd-content axis at all. It is a transfer check for the decomposition, not for the sign-flip phenomenon.
- The independent-evaluation task is small ($n = 7$) and its rectification gap is an order of magnitude smaller than the primary task's, so it is a transfer check, not a demonstration.
- If a hardware or publicly measured dataset showed the opposite sign for the benefit of orbit averaging at matched shots, the decomposition would survive but the operational reading would not; that is the experiment that would most directly refute the useful part of this paper.

Reproducibility

Code: `shotcert/` (`core.py`, `qsim.py`, `tasks.py`, `physics.py`, `certify.py`, `experiments/`). Tests: `python3 -m unittest discover -s shotcert -t . -p 'test_*.py'` (16 tests, including the decomposition identity, the feasibility properties, and non-degeneracy checks for every task). Results: `shotcert/results/{control,physics,variance}.json`, compiled into `shotcert/RESULTS.md` by `shotcert/compile_results.py`; every num-

Bound	S	n=25	n=50	n=100	n=200	n=400
eb	8	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.93/0.00
eb	32	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.99/0.00
eb	128	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
normal	8	0.10/0.00	0.28/0.00	0.48/0.00	0.87/0.00	0.99/0.00
normal	32	0.42/0.00	0.92/0.00	1.00/0.00	1.00/0.00	1.00/0.00
normal	128	0.82/0.00	1.00/0.00	1.00/0.00	1.00/0.00	1.00/0.00

Table 5: Power/violation rate for the distribution-free empirical-Bernstein bound and the tighter Studentised bound, averaging-helps regime, $\epsilon = 0$, $\delta = 0.05$. A bound is admissible only where its measured violation rate stays at or below δ.

Task	$R(a_0)$	best policy	R(best)	Δ	max resid.
heisenberg_even_target_n6	0.02167	strat	0.01367	−0.00800	2.8e-17
heisenberg_mostly_even_n6	0.01870	mix75	0.01387	−0.00483	2.8e-17
heisenberg_raw_target_n6	0.01621	mix25	0.01525	−0.00096	2.4e-17
heisenberg_even_target_n8	0.01734	strat	0.01319	−0.00415	1.2e-17
cluster_even_target_n7	0.01137	strat	0.00934	−0.00203	1.6e-17
cluster_raw_target_n7	0.01137	strat	0.00934	−0.00203	1.6e-17

Table 6: Physical tasks at $S = 64$ total shots, enumerated risk. The sign of the benefit of orbit averaging flips with the odd content of the target.

ber quoted here is emitted by that compiler. Protocol and novelty gate: `shotcert/docs/PROTOCOL.md`, `shotcert/docs/NOVELTY_MEMO.md`.

References

[1] Catak. Distribution-free generalization bounds for quantum measurement hypotheses with separate sample and shot budgets, and a closed-form allocation rule. arXiv:2608.24891, 2026.

[2] Chen. Statistical symmetry release: certified descent by relaxing equivariance. arXiv:2609.11470, 2026.

[3] Chen. Symmetry discovery: finite-measurement inference of symmetry groups. arXiv:2610.00157, 2026.

[4] Wang, et al. Fourier symmetrization for geometric quantum machine learning. arXiv:2610.01874, 2026.

[5] Cai, et al. Randomized symmetry in variational quantum algorithms. arXiv:2110.13116, 2021.